



Food Industry Development, Export Performance, and Sustainable Agricultural Value Added in Algeria: An ARDL Approach to Environmental and Food Production Policy

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Received: 03-26-2026

Revised: 05-23-2026

Accepted: 06-25-2026

Citation: Brika, S., Bengana, I., Khababa, N., & Adli, B. (2026). Food industry development, export performance, and sustainable agricultural value added in Algeria: An ARDL approach to environmental and food production policy. *Chall. Sustain.*, 14(5), 962–976. <https://doi.org/10.56578/cis140506>.



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Abstract: This study evaluates the contribution of the food and beverage industry and food and beverage exports toward sustainable agricultural development in Algeria. Considering the pressure for food security, resource scarcity, and the imperative for diversification away from hydrocarbons, the paper poses whether agro-industrial activities enhance the agricultural value added and addresses the food system resilience and sustainable development of the environment. The study analyzes the annual data of the National Office of Statistics of Algeria and uses the Autoregressive Distributed Lag (ARDL), bounds testing approach to analyze the short and long run relationships of agricultural value added, gross production of the food industries, and food and beverage exports. The findings affirm that there is a long-run equilibrium relationship among the stated variables. It is shown that in the long-run, a 1% increase in gross production of the food industries is associated with a 0.491% increase in agricultural value added, and a 1% increase in food and beverage exports would yield a 0.107% increase in agricultural value added. The error-correction coefficient showed that around 82% of short-run deviations from the long-run equilibrium are corrected within a year. This shows that the food-processing and export capacity may be associated with stronger the agricultural development of Algeria, sustainably framing agro-industrial growth, the reduction of post-harvest losses, and the environmentally sustainable food production policy.

Keywords: Food and beverage industry; Agricultural value added; Sustainable agricultural development; Food and beverage exports; ARDL bounds testing; Cointegration; Resource efficiency; Algeria

1. Introduction

The agricultural sector is imperative for Algeria's economy due to historical reasons concerning its importance to rural livelihoods, innervation of food supply, economic development, and environmental sustainability. Oil and gas affect Algeria's exports. Diversifying into agriculture can modernize an economically archaic sector to fulfil employment and food security in rural areas (Baghdad, 2022; Kahina, 2025). The trade-off is agricultural productivity and increasing the crop's ability to trade value with increasingly scarce resources such as fragmented land, inefficient systems of trade, and inadequate technologies of modern production and trade. In the context of improving the food industry, value addition to the first stage of agriculture and ensuring the farmers have a market, which lessens the losses in the food supply chain, will strengthen the food system and remove supply chain barriers (Mostefai, 2024).

Efficiency can be improved through agro-processing while bearing significance to environmental concerns. Value addition to agriculture will increase productivity and crop value, and fresh produce will endure the punishment of post-harvest. However, without a coordinated strategy to manage the burden of water, land, and resources with production methods that put a premium on efficiency, the food industry will take excessive toll on

the already strained resources (Bouznit & Aïssaoui, 2024). The burden of the food system for the agriculture sector is not an isolated problem, but a value addition to agriculture that merges the industry, environment, and food production.

Having a significant share in Algeria's GDP and exports, hydrocarbons dominate Algeria's economy. This reliance on a single sector has resulted in a lack of stability due to fluctuations in the price of oil and the global economy (Kahina, 2025). Thus, policymakers have named the food sector and agriculture, along with the food industry's capacity to add value to primary agricultural products and improve rural economic development, as areas that can help achieve agricultural transformation (Brika et al., 2021; Irandoust, 2022; Naili, 2014; Swearingen, 2019).

The food sector affects the development of agriculture in a variety of ways, but the most important are backward and forward linkages. Backward linkages occur in an industry when the industry procures primary products from local agricultural producers, which encourages local farmers to increase their agricultural output, as well as the adoption of modern agricultural practices (Hirschman, 1958). Conversely, forward linkages occur when value-added processing, packaging, and distribution of agricultural products with the aim of enhancing market accessibility, as well as improving the competitiveness of locally produced products in both domestic and international markets (Abdelhedi & Zouari, 2020; Porter, 1990). Other developing countries have taught us a lot about the importance of these linkages in improving agricultural productivity, as well as the incomes of people in rural areas (Lemaini, 2024).

The complexities of Algeria's structure and institutions influence the food industry's contribution to the development of agriculture. Land constrictions, water shortages, and the fragmentation of agricultural holdings, among other things, limit the country's agricultural productivity and efficiency. Smallholders with little to no access to agricultural inputs and technology are also high dominating. Nevertheless, the supply side also faces challenges; the sector's ability to respond to the 'growing food demand' is also limited (Urugo et al., 2024). All of these challenges are extremely complex and require integrating food systems into national agricultural development plans (Mostefai, 2024; Naili, 2014; Williams et al., 2020).

Recent years have seen several policy adjustments in Algeria regarding the agro-industrial sector as a new way to diversify the economy. These policy changes include the establishment of agro-industrial food processing plants, the provision of subsidies for the purchase of modern agricultural production tools, and the establishment of public-private partnerships (El Bilali & Ben Hassen, 2024). To some extent, the identified policy measures have demonstrated a positive outcome. However, the policies' usefulness in creating adequate linkages to the food industry is still debated among scholars (Benabbes & Laouar, 2021). However, the policies' contribution to agricultural development and the development of rural areas has been mostly unwritten (Boudkhil et al., 2020; Wahida, 2023).

Theoretical frameworks that analyze the intersection between the food industry and agriculture stream from certain elements like structural transformation and value chain development. Structural transformation is when resources are reallocated from low-productivity activities, like subsistence agriculture, to those with higher productivity, such as agro-industrial enterprises (Diao et al., 2010; Khalfa Brika et al., 2025). On the other hand, value chain development examines the integration of smallholder farmers into commercial supply chains and how this integration boosts their market access and income-generating opportunities (Kaplinsky & Morris, 2000). By utilizing the above-mentioned frameworks, one can understand how the food industry can catalyze the process of modernizing agriculture in Algeria (Hirschman, 1958).

Algeria can learn from other countries that have created a successful food industry. For example, Tunisia have a similar socioeconomic status, and showcasing them proves that with a little investment in the food industry (specifically food processing, food distribution, and food marketing), the agricultural industry booms and loses no crops (Moula & Djebala, 2025). Especially in an environment like Tunisia that fosters food industry innovations with created food industry and food processing infrastructure. Nonetheless, it is uncertain whether Algeria can mirror Tunisia because Algeria's economic issues are a lot more severe than those countries (Benmehaia & Bédrani, 2026).

While agriculture and the food industry in Algeria possess great potential for complementarity, the country faces logistical challenges, limited industrial processing in numerous regions, inadequate storage systems, and a lack of credit accessibility for smallholder farmers. Particularly, these obstacles restrict weak linkages between agro-industrial value chains and reduce the capabilities of the food processing industry to yield agricultural diversification (Mostefai, 2024). To overcome these challenges, an integrated policy framework with synergistic effect on market linkages, rural infrastructure, and agro-industrial growth, and resource sustainability is deemed necessary (Salah, 2025).

This study aims to assess the effect of the food sector on the advancement of the agricultural sector in Algeria using econometric techniques. It attempts to determine the main catalysts of agricultural growth and examines the contribution of the food sector by evaluating the time series of agricultural outputs, the food sector, and other pertinent economic variables. It is believed that the results will assist policymakers and other stakeholders in understanding how the food sector can be leveraged to promote the sustainable advancement of the agricultural

sector (Abdelhedi & Zouari, 2020).

This study contributes to the literature in three ways. First, it provides Algeria-specific time-series evidence on the relationship between food industry output, food and beverage exports, and agricultural value added. Second, it distinguishes between short-run dynamics and long-run equilibrium effects using the Autoregressive Distributed Lag (ARDL) bounds testing approach. Third, it interprets the estimated relationships through the lens of sustainable agricultural development, emphasizing that agro-industrial expansion must be assessed in relation to resource efficiency, post-harvest loss reduction, and food-system resilience.

The remainder of the study consists of the following sections: Section 2 reviews the literature on the interaction between the food industry and agricultural development. Sections 3 and 4 detail the data and methods used in the study. Section 5 presents the results. Section 6 discusses the findings, offers insights, and suggests practical policy recommendations for further research.

2. Literature Review

Ample research has been done on the intertwining of the food industry and agricultural development. Researchers have noted the potential benefits of a well-developed agro-industrial sector. Hirschman (1958) stated that for economic development to occur, backward and forward linkages must be formed where industries develop demand for a primary product and in turn, boost agricultural production. In light of the structural obstacles presented for Algeria's agricultural sector, understanding these linkages is vital (Benmehaia & Bédrani, 2026).

Value chain analysis is a key factor of agricultural development, and this has been studied by Kaplinsky & Morris (2000). They detailed how the food industry aids in the marketing of smallholder farmers by cutting down transaction costs and providing a consistent need for agricultural products. This is in line with Lemaini (2024) where the authors have shown that the construction of food processing facilities leads to a higher agricultural productivity and enhanced rural income.

Recent studies have revealed that the food industry can act as an impetus for the modernization of agriculture in North Africa. Anyanwu & Kponnou (2017) study of agro-industrial investments in North Africa showed that these investments boost agricultural productivity and reduce post-harvest losses. They studied an area that is relevant to Algeria, as handling losses after harvesting is an area that remains unproductive in Algeria (Wahida, 2023).

Johnson et al. (2022) studied the food industry role in rural economies. They reported that construction of food processing units in developing nations leads to a creation of jobs and development of rural economies. This construction of food processing units in developing nations leads to a development of rural economies. Kahina (2025) is of the same opinion, as he supports the development of agro-industrial units in Algeria to reduce the dependency on hydrocarbons.

The food industry aids food security not only by spurring economic growth but also by increasing the range and decreasing the cost of processed food items. Supply chain stabilisation also means stronger food industries can mitigate concerns regarding the risk of fluctuating agricultural yields (El Bilali & Ben Hassen, 2024; Irandoust, 2022). This is especially the case in Algeria where unstable climate conditions affect the state of agricultural productivity (Benmehaia & Bédrani, 2026).

The study of the relationship between the food industry and agriculture has largely been based on the concept of structural transformation. Diao et al. (2010) posited that the shifting of resources from subsistence agriculture to agro-industrial companies is one of the primary indicators of an economy's progression. This is also the case in Porter (1990) where the description of competitive advantage in terms of economic growth provides an explanation for control in various economic sectors and the quick development of inter-sectoral relationships.

Although the aforementioned theories exist, the potential interrelations of the food industry and agriculture in Algeria still face various challenges. Williams et al. (2020) cited inadequate infrastructure, insufficient funding, and poor logistical systems as the primary challenges. These challenges can only be addressed by focusing policies and investments on these areas (Benabbes & Laouar, 2021). The literature highlights the importance of the food industry in developing the agriculture sector. The application of such literature to Algeria, however, remains an issue. The study attempts to fill this void by exploring the food industry-agriculture dynamics in Algeria in an attempt to identify the main variables of the food industry and the inherent policy options.

Although previous studies have examined agricultural development, food security, and agro-industrial transformation in Algeria and North Africa, four limitations remain. First, many contributions rely on descriptive or policy-based evidence rather than formal time-series estimation. Second, the literature rarely distinguishes between short-run adjustment and long-run equilibrium relationships linking food industry activity to agricultural value added. Third, food and beverage exports are often absorbed into broader trade indicators rather than treated as a distinct agro-industrial channel. Finally, the environmental implications of agro-industrial expansion, including water pressure, land fragmentation, supply-chain efficiency, and post-harvest loss reduction, remain underdeveloped in Algeria-specific empirical research. The present study addresses these gaps by applying an ARDL framework to Algeria and by interpreting the results within environmental economics and sustainable food

production policy (Ferrah & Oubelli, 2013; Kahina, 2025; Mostefai, 2024).

3. Data

The empirical analysis is based on annual time-series data covering the period from 1987 to 2021, resulting in a final sample of 35 annual observations for each variable. The data were obtained from the National Office of Statistics. The study examines the relationship between the development of the agricultural sector, measured by agricultural value added, and selected indicators related to the food-processing industry and food trade. Specifically, the dependent variable is agricultural value added, while the explanatory variables represent the gross production of food industries and exports of food products and beverages.

Prior to the econometric analysis, the data were carefully reviewed and prepared to ensure consistency across the observation period. The series were checked for missing observations, consistency of units and definitions, and chronological continuity. The variables were then transformed into natural logarithms in order to reduce scale differences, mitigate potential heteroscedasticity, and allow the estimated coefficients to be interpreted in relative rather than absolute terms. The resulting logarithmic series were subsequently used in the time-series econometric procedures, including the stationarity and ARDL analyses.

The dependent variable, $LVAAG$, denotes the natural logarithm of agricultural value added. Agricultural value added represents the contribution of the agricultural sector to overall economic activity after accounting for the value of intermediate inputs used in agricultural production. It therefore provides an economic measure of the size and performance of the agricultural sector. The variable is measured in monetary units and is transformed as follows:

$$LVAAG_t = \ln (VAAG_t)$$

The first explanatory variable, $LPBIA$, represents the natural logarithm of the gross production of food industries. This variable captures the level of production generated by food-processing activities and is therefore used as an indicator of the scale of the food-processing industry and its potential contribution to agricultural development through the demand for agricultural raw materials. It is expressed in monetary units and is transformed according to:

$$LPBIA_t = \ln (PBIA_t)$$

The second explanatory variable, $LEXAB$, denotes the natural logarithm of exports of food products and beverages. This variable reflects the external demand for food-industry products and provides an indicator of the integration of food-related activities into international markets. It is measured in monetary units and is transformed as follows:

$$LEXAB_t = \ln (EXAB_t)$$

The logarithmic specification also changes the interpretation of the estimated coefficients. In a model in which the dependent variable and an explanatory variable are both expressed in natural logarithms, the corresponding coefficient can be interpreted as an elasticity. Thus, a coefficient associated with $LPBIA$, for example, measures the percentage change in agricultural value added associated with a 1% change in the gross production of food industries, *ceteris paribus*. Similarly, the coefficient of $LEXAB$ represents the percentage change in agricultural value added associated with a 1% change in food-product and beverage exports, holding the other explanatory variables constant.

Accordingly, the logarithmic transformation does not merely constitute a mathematical rescaling of the original observations; it provides an economically meaningful specification in which the estimated relationships can be interpreted in terms of proportional or percentage changes. This interpretation is particularly appropriate for macroeconomic time-series variables whose magnitudes may differ substantially across indicators and whose relationships are more meaningfully expressed in relative terms.

4. Methodology

4.1 Definition and Advantages of Autoregressive Distributed Lag Model

The ARDL model, introduced by Pesaran et al. (2001), is a robust econometric technique designed to test for cointegration and to estimate both short- and long-term relationships between variables. This method is particularly valuable when analyzing the interplay of variables in time-series data, regardless of their order of integration.

4.2 General Form of the Autoregressive Distributed Lag Model

The general ARDL model, denoted as $ARDL(p, q_1, q_2)$, for a dependent variable y_t and two explanatory variables, x_{1t} and x_{2t} , can be expressed as:

$$y_t = \alpha_0 + \sum_{i=1}^p \alpha_i y_{t-i} + \sum_{j=0}^{q_1} \beta_j x_{1t-j} + \sum_{k=0}^{q_2} \gamma_k x_{2t-k} + \varepsilon_t$$

where, y_t denotes the dependent variable at time t . In this study, it represents *LVAAG*, the natural logarithm of agricultural value added. x_{1t} denotes the first explanatory variable at time t . In this study, it represents *LPBIA*, the natural logarithm of the gross production of food industries. x_{2t} denotes the second explanatory variable at time t . In this study, it represents *LEXAB*, the natural logarithm of exports of food products and beverages. α_0 denotes the constant (intercept) term of the model. α_i denotes the autoregressive coefficients associated with the lagged values of the dependent variable y_{t-i} , for $i = 1, \dots, p$. β_j denotes the coefficients associated with the current and lagged values of the first explanatory variable x_{1t-j} , for $j = 0, \dots, q_1$. γ_k denotes the coefficients associated with the current and lagged values of the second explanatory variable x_{2t-k} , for $k = 0, \dots, q_2$. p denotes the lag order of the dependent variable y_t . q_1 denotes the lag order of the first explanatory variable x_{1t} . q_2 denotes the lag order of the second explanatory variable x_{2t} . ε_t denotes the error term (white noise disturbance) at time t , assumed to be independently and identically distributed with zero mean and constant variance.

4.3 Unrestricted Error Correction Model Derivation

The ARDL model can be reformulated into an unrestricted Error Correction Model (ECM), which is used to test for cointegration (Bounds Test) and estimate the relationship between variables. The reformulated model is expressed as:

$$\Delta y_t = \beta_0 + \sum_{i=1}^p \beta_i \Delta y_{t-i} + \sum_{j=0}^{q_1} \gamma_j \Delta x_{1t-j} + \sum_{k=0}^{q_2} \delta_k \Delta x_{2t-k} + \theta_1 y_{t-1} + \theta_2 x_{1t-1} + \theta_3 x_{2t-1} + \varepsilon_t$$

where, Δy_t denotes the first difference of the dependent variable at time t , i.e., $\Delta y_t = y_t - y_{t-1}$. It represents the short-run change in *LVAAG*. Δy_{t-i} denotes the lagged first differences of the dependent variable, for $i = 1, \dots, p$. Δx_{1t-j} denotes the current and lagged first differences of the first explanatory variable, for $j = 0, \dots, q_1$. It represents short-run changes in *LPBIA*. Δx_{2t-k} denotes the current and lagged first differences of the second explanatory variable, for $k = 0, \dots, q_2$. It represents short-run changes in *LEXAB*. y_{t-1} denotes the one-period lagged level of the dependent variable. x_{1t-1} denotes the one-period lagged level of the first explanatory variable. x_{2t-1} denotes the one-period lagged level of the second explanatory variable. β_0 denotes the constant (intercept) term in the ECM. β_i denotes the short-run coefficients associated with the lagged first differences of the dependent variable. γ_j denotes the short-run coefficients associated with the current and lagged first differences of the first explanatory variable. δ_k denotes the short-run coefficients associated with the current and lagged first differences of the second explanatory variable. θ_1 , θ_2 , and θ_3 denote the long-run parameters associated with the lagged level terms y_{t-1} , x_{1t-1} , and x_{2t-1} , respectively. These are used in the Bounds Test to determine whether a long-run (cointegrating) relationship exists among the variables. ε_t denotes the error term at time t , assumed to be white noise. θ_1 , θ_2 , and θ_3 represent the long-term parameters. β_i , γ_j , and δ_k denote the short-term parameters.

The Bounds Test is applied to determine whether a long-term relationship exists among the variables. If cointegration is confirmed, the unrestricted ECM is used to estimate both short- and long-term dynamics.

The ARDL model offers several advantages, making it a widely used econometric approach for analyzing time-series data. Its flexibility and robustness in handling various integration orders and small sample sizes have established it as a preferred technique in applied econometrics.

Firstly, one of the key advantages of ARDL models is their ability to handle variables with different integration orders. Unlike other cointegration methods, such as the Johansen or Engle-Granger approaches, the ARDL model does not require all variables to be integrated of the same order. It can accommodate a mix of stationary (I(0)) and first-order integrated (I(1)) variables, making it versatile for empirical research (Pesaran et al., 2001). This feature allows researchers to avoid the challenges associated with pre-testing for unit roots, which can often lead to errors in model specification.

Secondly, ARDL models are particularly effective in studies with small sample sizes. Traditional cointegration techniques often require large datasets to produce reliable results, but the ARDL approach performs well even with limited observations (Narayan & Smyth, 2005). This makes it an invaluable tool for studies in fields where

data availability is constrained, such as regional or sector-specific analyses.

Another notable advantage is the ability of the ARDL framework to estimate short-run and long-run relationships simultaneously. The model separates the dynamics of short-term adjustments from long-term equilibrium relationships within a single estimation process, providing a comprehensive understanding of variable interactions (Pesaran et al., 2001). This dual focus enhances the interpretability of results and offers more actionable insights for policymakers and researchers.

Additionally, the ARDL model incorporates lag structures for both dependent and independent variables. By accounting for the lagged effects of variables, it captures the temporal dynamics of relationships more effectively than models that rely solely on contemporaneous relationships (Tenkir, 2022). This feature is particularly useful for economic and financial analyses, where delayed responses to policy changes or shocks are common.

Finally, the ARDL approach allows for the use of the Bounds Test to determine cointegration. This test is robust and simple to implement, providing clear criteria for the existence of long-term relationships without requiring knowledge of the exact integration orders of the variables (Pesaran et al., 2001). The ability to perform cointegration testing alongside model estimation adds to the methodological efficiency of ARDL.

In summary, the ARDL model offers significant advantages, including its flexibility in handling mixed integration orders, suitability for small sample sizes, simultaneous estimation of short-run and long-run relationships, inclusion of lagged effects, and robust cointegration testing via the Bounds Test. These features make it an indispensable tool for empirical research in economics and other social sciences.

5. Results and Discussion

5.1 Analysis of the Relationship Between Variables

5.1.1 Study variables

The representation of the study variables in Figure 1 reveals that they exhibit similar trends and move in the same direction over time. This visual alignment suggests the presence of a potential long-term equilibrium relationship between the variables, which warrants further investigation. Such co-movement is often indicative of underlying economic linkages, particularly in sectors like agriculture and food production, where interdependencies are common (Allaoua & Krelifa, 2026).

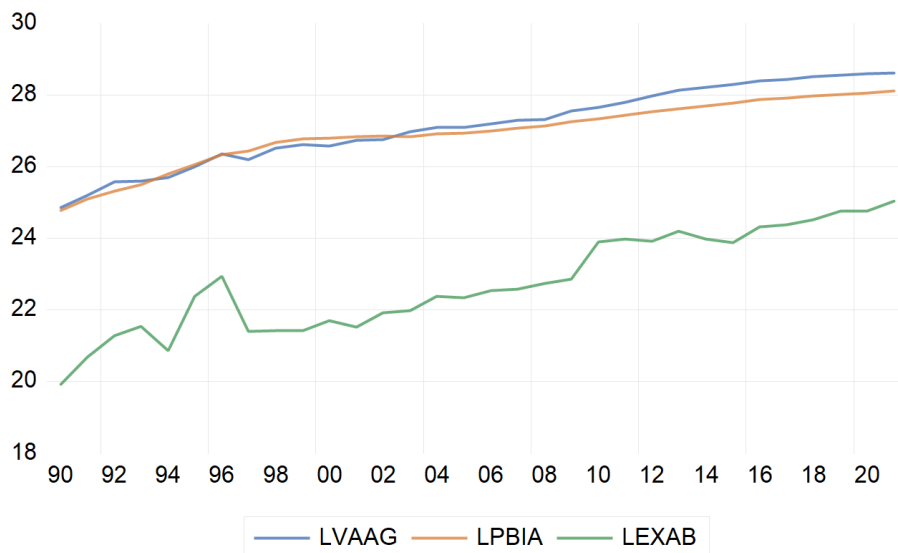


Figure 1. Joint graphical representation of study variables

Note: *LVAAG*, natural logarithm of agricultural value added; *LPBIA*, natural logarithm of the gross production of food industries; *LEXAB*, natural logarithm of exports of food products and beverages. Source: Prepared by the authors (*EViews 13 Output*).

5.1.2 Nature of the relationship between variables

Figure 2 indicates that the relationship between the variables can be modeled as a linear relationship. This observation aligns with economic theory, which often assumes linear dependencies between sectoral outputs and their contributing factors. However, to confirm this linearity, a formal correlation analysis was conducted.

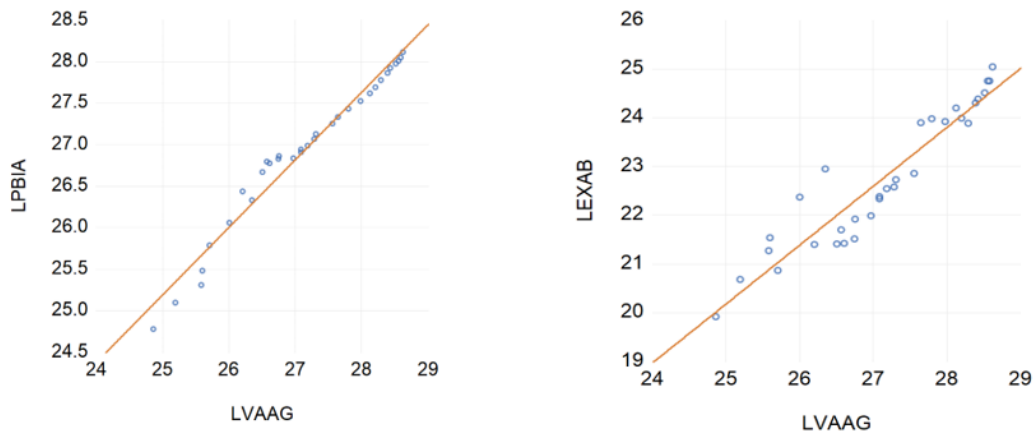


Figure 2. Relationship between variables

Note: *LVAAG*, natural logarithm of agricultural value added; *LPBIA*, natural logarithm of the gross production of food industries; *LEXAB*, natural logarithm of exports of food products and beverages. Source: Prepared by the authors (*EViews 13 Output*).

5.1.3 Correlation analysis between variables

To quantify the strength and direction of the relationship between the variables, the correlation coefficient was calculated. The results are presented in Table 1.

Table 1. Correlation matrix of study variables

Variables	<i>LVAAG</i>	<i>LPBIA</i>	<i>LEXAB</i>
<i>LVAAG</i>	1.000	0.983	0.943
<i>LPBIA</i>	0.983	1.000	0.897
<i>LEXAB</i>	0.943	0.897	1.000

Note: *LVAAG*, natural logarithm of agricultural value added; *LPBIA*, natural logarithm of the gross production of food industries; *LEXAB*, natural logarithm of exports of food products and beverages. Source: Prepared by the authors (*EViews 13 Output*).

The correlation matrix reveals a strong positive correlation between the study variables, with correlation coefficients close to 1. Specifically, *LVAAG* shows a robust direct relationship with *LPBIA* and *LEXAB*. These findings are consistent with prior studies that highlight the interconnectedness of agricultural output and food industry performance (Bouznit & Aïssaoui, 2024; Chaib, 2022; Montalbano & Nenci, 2022; Tambe et al., 2023). The strong correlations provide preliminary evidence of a significant relationship, which was further investigated using regression analysis.

5.1.4 Estimation of the regression model

The regression model was estimated to examine the relationship between the dependent variable, *LVAAG*, and the independent variables, *LPBIA* and *LEXAB*. The results are summarized in Table 2.

Table 2. Regression model estimation

OLS Estimation Results—Dependent Variable: <i>LVAAG</i>				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-1.336	0.923	-1.447	0.158
<i>LPBIA</i>	0.853	0.060	14.05	<0.001
<i>LEXAB</i>	0.241	0.039	6.147	<0.001
Regression Summary Statistics and Diagnostic Measures				
Statistic	Value	Statistic	Value	
R-squared	0.987	Mean dependent var.	27.136	
Adjusted R-squared	0.985	S.D. dependent var.	1.084	
S.E. of regression	0.134	Akaike info. criterion	-1.095	
Sum squared resid	0.520	Schwarz criterion	-0.957	
Log likelihood	20.517	Hannan-Quinn criterion	-1.049	
F-statistic	1002.629	Durbin-Watson statistic	0.764	
Prob(F-statistic)	<0.001			

Note: OLS, ordinary least squares, C, constant (intercept) term; *LVAAG*, natural logarithm of agricultural value added; *LPBIA*, natural logarithm of the gross production of food industries; *LEXAB*, natural logarithm of exports of food products and beverages. Source: Prepared by the authors (*EViews 13 Output*).

The Durbin-Watson statistic (0.764) is well below the conventional lower bound, indicating the presence of strong positive autocorrelation in the residuals. This raises concerns about the reliability of the regression estimates, as autocorrelation can lead to inefficient estimates and biased standard errors, potentially producing spurious results (Wooldridge, 2016).

5.1.5 Unit root tests and robustness checks

To examine the stationarity properties of the time series, the Augmented Dickey–Fuller (ADF) test was applied to all variables in the model under three alternative deterministic specifications: trend and intercept, intercept only, and without trend and intercept. The results are reported in Table 3. At the level form, the ADF results provide evidence of non-stationarity for *LVAAG* across the three specifications. For *LPBIA* and *LEXAB*, however, the results vary across the deterministic specifications, with the null hypothesis of a unit root being rejected under some specifications but not under others. After first differencing, *LEXAB* is stationary under all three specifications, whereas *LVAAG* and *LPBIA* show mixed results depending on the deterministic specification. These findings indicate that the stationarity results should be interpreted in relation to the appropriate deterministic specification for each series.

Table 3. Augmented Dickey–Fuller (ADF) unit root test results

Variable	At Level			First Difference		
	Trend & Intercept	Intercept Only	None	Trend & Intercept	Intercept Only	None
<i>LVAAG</i>	0.777	0.355	0.988	<0.001***	<0.001***	0.081
<i>LPBIA</i>	<0.001***	0.498	0.990	0.157	0.016*	0.017*
<i>LEXAB</i>	0.049*	0.552	0.976	<0.001***	<0.001***	<0.001***

Note: *LVAAG*, natural logarithm of agricultural value added; *LPBIA*, natural logarithm of the gross production of food industries; *LEXAB*, natural logarithm of exports of food products and beverages. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. “None” denotes the specification without trend and intercept. p -values below 0.001 are reported as $p < 0.001$ rather than 0.000. Source: Prepared by the authors (*EViews 13 Output*).

5.1.6 Granger predictive causality tests

To examine the predictive direction among the study variables, the Granger predictive causality test was conducted using three lags (Lag = 3), as selected by the vector autoregression (VAR) lag-order selection criteria. The results of the lag-order selection are reported in Table 4, the results of the Granger causality test between the study variables are presented in Table 5.

Table 4. Vector autoregression (VAR) lag order selection criteria

Lag	LogL	LR	FPE	AIC	SC	HQ
0	47.820	NA	0.000007	-3.320	-3.176	-3.277
1	67.107	32.860	0.000003	-4.082	-3.506	-3.911
2	76.401	13.768	0.000003	-4.104	-3.096	-3.804
3	99.279	28.810*	0.000001*	-5.132*	-3.692*	-4.704*
4	107.848	8.886	0.000002	-5.100	-3.228	-4.543

Note: LR, Likelihood Ratio test statistic; FPE, Final Prediction Error criterion; AIC, Akaike Information Criterion; SC, Schwarz Information Criterion; HQ, Hannan–Quinn Information Criterion. Endogenous variables: *D(LVAAG)*, *D(LPBI)*, *D(LEXAB)*. Source: Prepared by the authors (*EViews13 Output*).

Table 5. Granger causality test between variables

Null Hypothesis	Obs.	F-Statistic	Prob.
<i>D(LPBI)</i> does not Granger Cause <i>D(LVAAG)</i>	28	3.612	0.030*
<i>D(LVAAG)</i> does not Granger Cause <i>D(LPBI)</i>	28	3.771	0.026*
<i>D(LEXAB)</i> does not Granger Cause <i>D(LVAAG)</i>	28	2.863	0.061
<i>D(LVAAG)</i> does not Granger Cause <i>D(LEXAB)</i>	28	1.903	0.160
<i>D(LEXAB)</i> does not Granger Cause <i>D(LPBI)</i>	28	5.789	0.005**
<i>D(LPBI)</i> does not Granger Cause <i>D(LEXAB)</i>	28	3.723	0.027*

Note: *D(.)* denotes the first difference of the variable. *LVAAG*, natural logarithm of agricultural value added; *LPBIA*, natural logarithm of the gross production of food industries; *LEXAB*, natural logarithm of exports of food products and beverages. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Source: Prepared by the authors (*EViews13 Output*).

The results indicate a bidirectional Granger predictive relationship between changes in *LPBIA* and *LVAAG* at the 5% significance level. Specifically, changes in *LPBIA* help predict subsequent changes in *LVAAG* ($p = 0.030$), while changes in *LVAAG* also help predict subsequent changes in *LPBIA* ($p = 0.026$). In contrast, the predictive relationship from *LEXAB* to *LVAAG* is not statistically significant at the 5% level ($p = 0.061$), nor is the relationship from *LVAAG* to *LEXAB* ($p = 0.160$). However, the results indicate a bidirectional Granger predictive relationship

between *LEXAB* and *LPBIA*, with *p*-values of 0.005 and 0.027, respectively. These findings indicate predictive precedence among some of the variables but should not be interpreted as definitive evidence of structural or economic causality.

5.1.7 Integration, model selection, and Autoregressive Distributed Lag estimation

Integration: The unit root analysis indicates that the study variables are not integrated beyond the first order; in particular, none of the series exhibits evidence of being integrated of order two ($I(2)$) or higher. The results of the ADF tests, considered under the appropriate deterministic specifications, provide the basis for proceeding with the cointegration analysis.

Causality: The Granger causality analysis provides evidence regarding the direction of predictive causality among the study variables. These results, together with the integration analysis, provide a basis for proceeding to the subsequent analysis of the dynamic relationships among the variables.

Next Step: Based on the unit root and Granger causality results, the relationships among the study variables are subsequently examined using the ARDL approach. The ARDL framework is appropriate when the variables are a mixture of $I(0)$ and $I(1)$, provided that none of the variables is $I(2)$ or higher. It also allows the estimation and interpretation of both long-run and short-run dynamics within a unified framework.

The integration analysis confirms that no variable is $I(2)$, while the predictive causality results indicate that food industry production and food and beverage exports precede movements in agricultural value added. These results justify the use of ARDL to estimate both long-run and short-run relationships among the study variables.

Based on the results of the Granger causality test and the integration analysis, the relationship between the study variables will be modeled using the ARDL approach in the next section.

5.2 Autoregressive Distributed Lag Model Estimation

After confirming that the variables are integrated of order one and that no series is integrated of order two or higher, the ARDL model was employed to estimate the relationship. The optimal lag structure was determined using the Akaike Information Criterion (AIC), as illustrated in Figure 3.

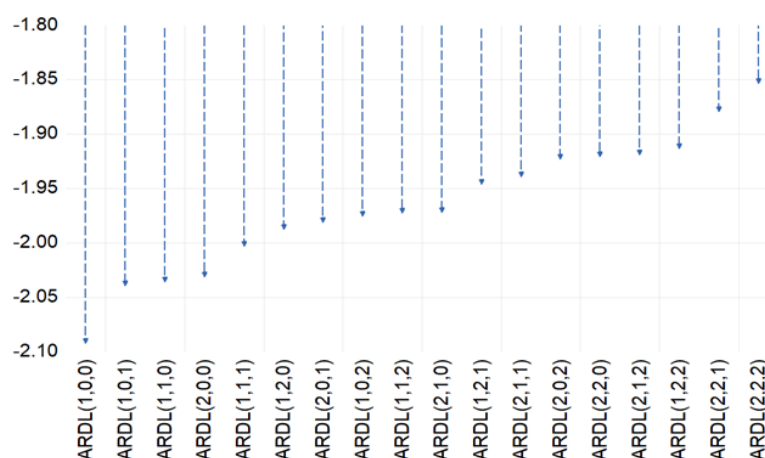


Figure 3. Optimal Autoregressive Distributed Lag (ARDL) model selection based on the Akaike criterion

Source: Prepared by the authors (*EViews13 Output*).

The analysis identifies ARDL(1,0,0) as the best model, which was subsequently used for cointegration testing and long-term relationship estimation.

5.2.1 Bounds test for cointegration

The bounds test was conducted to determine the existence of a long-term equilibrium relationship between the variables. The results are presented in Table 6.

The computed *F*-statistic (12.175) exceeds the upper bound critical values at all significance levels, leading to the rejection of the null hypothesis. This confirms the presence of a long-term equilibrium relationship between the study variables. The *t*-statistic ($t = -5.140$) further supports this conclusion, consistent with the findings of similar studies in developing economies (Keppler, 2007).

5.2.2 Long-term relationship estimation

The long-term relationship between the variables was estimated using the ARDL model. The results are

summarized in Table 7.

Table 6. Bounds test results

Test Statistic	Value	Significance	I(0)	I(1)
<i>F</i> -statistic	12.175	10%	4.190	5.060
<i>k</i>	2	5%	4.870	5.850
		2.5%	5.790	6.590
		1%	6.340	7.520
Actual sample size	31	Finite sample: <i>n</i> = 35		
		10%	4.517	5.480
		5%	5.457	6.570
		1%	7.643	9.063
		Finite sample: <i>n</i> = 30		
		10%	4.577	5.600
		5%	5.550	6.747
		1%	7.977	9.413
Panel B: <i>t</i>-Bounds test				
<i>t</i> -statistic	-5.140	10%	-3.130	-3.630
		5%	-3.410	-3.950
		2.5%	-3.650	-4.20
		1%	-3.960	-4.530

Note: *k*, number of explanatory variables in the ARDL model (here, *k* = 2). I(0), lower bound critical values assuming all variables are integrated of order zero, I(1), upper bound critical values assuming all variables are integrated of order one, I(1). Actual sample size, the number of observations used in the estimation (31). Finite sample: *n* = 35 and Finite sample: *n* = 30, critical values for finite sample sizes as reported by the software. Source: Prepared by the authors (*EViews13 Output*).

Table 7. Long-term relationship estimation (dependent variable: *LVAAG*)

Variable	Coefficient	Std. Error	<i>t</i> -Statistic	Prob.
<i>LPBIA</i>	0.491	0.082	5.956	<0.001***
<i>LEXAB</i>	0.107	0.043	2.484	0.020*

Note: *LVAAG*, natural logarithm of agricultural value added; *LPBIA*, natural logarithm of the gross production of food industries; *LEXAB*, natural logarithm of exports of food products and beverages. * *p* < 0.05, ** *p* < 0.01, *** *p* < 0.001. Source: Prepared by the authors (*EViews13 Output*).

LPBIA has a statistically significant and positive long-run relationship with agricultural value added (*p* < 0.05). The agricultural value added is estimated to increase by 0.491% with a 1% increase in food industry production. With *LEXAB*, a 1% increase in food and beverage exports results an estimated 0.107% increase in agricultural value added. The results show that food-processing and export activities can contribute to the agricultural sector, and the effects will be determined by the available resources, infrastructure, and organization of the value chain.

5.2.3 Short-term relationship estimation

The short-term dynamics of the relationship were analyzed using the ECM. The results are presented in Table 8.

Table 8. Short-term relationship estimation

Variable	Coefficient	Std. Error	<i>t</i> -Statistic	Prob.
C	8.804	1.372	6.418	<0.001***
@TREND	0.044	0.008	5.514	<0.001***
CointEq(-1)*	-0.823	0.131	-6.272	<0.001***

Note: C, constant (intercept) term; @TREND, deterministic time trend variable; CointEq(-1), one-period lagged error correction term, also known as the cointegrating equation residual. * *p* < 0.05, ** *p* < 0.01, *** *p* < 0.001. Source: Prepared by the authors (*EViews13 Output*).

The negative value and statistical significance of the error correction term supports the movement toward the long-run equilibrium. The coefficient of -0.823 implies that 82.3% of short-run equilibrium disequilibrium is corrected within the year. This relatively rapid adjustment suggests that value added in agriculture responds strongly to short-run deviations from the long-run trajectory of agriculture, the food industry, and the food & beverage industry exports.

These results are consistent with the findings of prior studies, which emphasize the role of error correction mechanisms in maintaining equilibrium in economic systems (Harris & Sollis, 2003).

5.3 Diagnostic Tests for the Model

To confirm the results of the estimated model, several diagnostic tests were conducted.

5.3.1 Autocorrelation test (Lagrange Multiplier test for errors)

The probability value (p -value) is 0.302, which is greater than the significance level of 5%. Therefore, we accept the null hypothesis (H_0), which states that there is no autocorrelation among the errors (Table 9).

Table 9. Breusch–Godfrey serial correlation Lagrange Multiplier (LM) test results

Test Statistic	Value	Prob.
F -statistic	1.004	0.381
Obs* R -squared	2.394	0.302

Source: Prepared by the authors (*EViews13 Output*).

5.3.2 Normality test for errors (Jarque-Bera test)

The Jarque-Bera test statistic is 1.076 (Table 10), and the probability value (p -value) is 0.584, which is greater than 0.05. Thus, we accept the null hypothesis (H_0), indicating that the errors follow a normal distribution.

Table 10. Normality test results

Statistic	Value
Skewness	-0.254
Kurtosis	2.242
Jarque-Bera	1.076
Prob.	0.584
Observations	31
Sample	1991–2021

Source: Prepared by the authors (*EViews13 Output*).

5.3.3 Heteroscedasticity test (Autoregressive Conditional Heteroskedasticity test)

The probability value (p -value) is 0.717, which is greater than 0.05. Therefore, we accept the null hypothesis (H_0), which states that the model does not suffer from heteroscedasticity, meaning the variance of the errors is homoscedastic (Table 11).

Table 11. Heteroscedasticity test results

Test Statistic	Value	Prob.
F -statistic	0.123	0.728
Obs* R -squared	0.132	0.717

Source: Prepared by the authors (*EViews13 Output*).

5.3.4 Model specification test (Ramsey Regression Equation Specification Error Test)

The probability value (p -value) is 0.190, which is greater than $\alpha = 0.05$. Thus, we accept the null hypothesis, indicating that there is no issue of model misspecification (the model is correctly specified) (Table 12).

Table 12. Ramsey Regression Equation Specification Error Test (RESET) results

Test Statistic	Value	df	Prob.
t -statistic	1.349	25	0.190
F -statistic	1.819	(1, 25)	0.190
Likelihood ratio	2.177	1	0.140

Note: df , degrees of freedom. Source: Prepared by the authors (*EViews13 Output*).

5.3.5 Structural stability test (Cumulative Sum of Recursive Residuals and Cumulative Sum of Squares of Recursive Residuals graphs)

The CUSUM and CUSUMSQ graphs (Figure 4) show no evidence of structural breaks, as the curves remain within the confidence bounds at the 5% significance level ($\alpha = 0.05$).

5.3.6 Diagnostic tests of the model

To assess the adequacy and stability of the estimated ARDL model, several diagnostic and stability tests were

conducted, as summarized in Table 13.

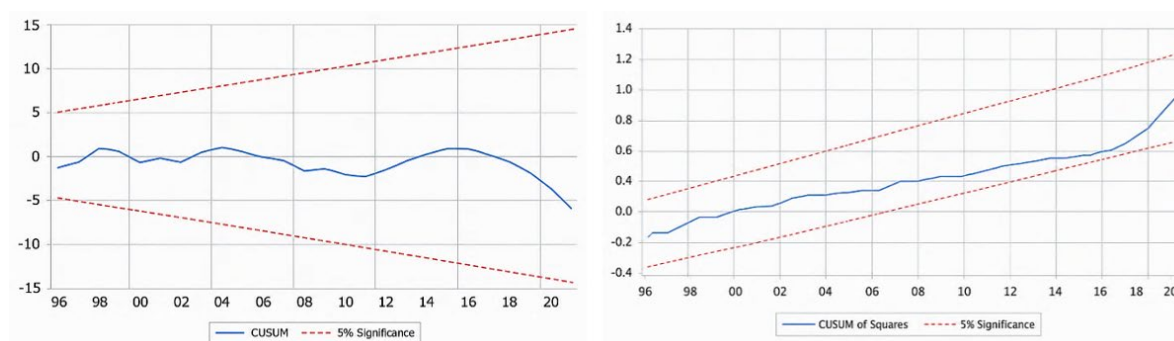


Figure 4. Graphical representation of CUSUM and CUSUMSQ

Note: CUSUM, Cumulative Sum of Recursive Residuals; CUSUMSQ, Cumulative Sum of Squares of Recursive Residuals. Source: Prepared by the authors (*EViews13 Output*).

Table 13. Summary of model diagnostic tests

Test	Method	Prob.	Conclusion
Autocorrelation	LM Test	0.302	No autocorrelation
Heteroscedasticity	ARCH Test	0.717	Homoscedasticity confirmed
Normality of Errors	Jarque-Bera Test	0.584	Errors follow normal distribution
Model Specification	Ramsey RESET Test	0.190	Model is correctly specified
Structural Stability	CUSUM & CUSUMSQ	Within bounds	No structural instability

Note: LM, Lagrange Multiplier; ARCH, Autoregressive Conditional Heteroskedasticity; RESET, Regression Equation Specification Error Test; CUSUM, Cumulative Sum of Recursive Residuals; CUSUMSQ, Cumulative Sum of Squares of Recursive Residuals. Source: Prepared by the authors (*EViews13 Output*).

The diagnostic results provide no evidence of major econometric problems in the estimated ARDL model. Specifically, the Lagrange Multiplier test indicates no statistically significant evidence of residual autocorrelation ($p = 0.302$), while the ARCH test provides no evidence of heteroscedasticity ($p = 0.717$). The Jarque–Bera test indicates that the residuals are consistent with normality ($p = 0.584$). In addition, the Ramsey RESET test provides no statistically significant evidence of model misspecification ($p = 0.190$). Finally, the CUSUM and CUSUMSQ tests indicate that the model remains structurally stable over the study period, as the statistics remain within the corresponding critical bounds. Taken together, these diagnostic and stability results support the adequacy and stability of the estimated ARDL specification.

6. Conclusion

The empirical data confirm an enduring equilibrium phenomenon among the production of the food industry, food and beverage exports, and the added value of agriculture in Algeria. The positive long-run coefficients indicate that the agro-industrial sector may foster agricultural development via backward and forward linkages. Backward linkages are concerned with growing trade of unprocessed agricultural products, while forward linkages are concerned with agro-industrial investments in processing, storage, packaging, and market infrastructure. However, the findings need to be interpreted in Algeria's structural context, which consists of water scarcity, fragmented land, poor storage facilities, and weak logistical infrastructure, which limit the expansion of agriculture.

From the perspective of environmental economics, the growth of the food industry has a mixed impact. On the positive side, if the level of processing is increased, post-harvest losses may be reduced, the agricultural raw material may be effectively utilized, and the economic returns from the output may be increased. On the negative side, if agro-industrial expansion occurs through production that is water-intensive, coupled with inefficient and poorly regulated supply irrigation, the pressure on the country's natural resource may increase. The positive link between food industry output and agricultural value added should be interpreted as the need to control agro-industrial expansion and integrate agro-industrial policy with other dimensions, such as the sustainable management of water, the efficient use of resources in production, and the resilience of food systems.

In the short term, the significant and negative error-correction term suggests that deviations from long-run equilibrium are corrected rapidly, which indicates the existence of adjustment mechanisms among agro-industrial activity and agricultural value added. While this establishes the need for careful policy making, especially in the context of environmental and resource limitations, it also underscores the importance of designing prudent policies in the face of environmental and institutional limitations.

The diagnostic test results indicate model reliability, as residual diagnostics reveal no serious autocorrelation,

heteroscedasticity, non-normality, or residual test issues. Diagnostic tests increase confidence in the specified model, while the theory of robustness notes that future research should implement broader boundary constraints as the data allow.

Policy implications are based on three empirical findings. First, positive related coefficients of gross production of food industries point toward stronger integration of food processing firms and domestic farmers using local, contract-based procurement, rural storage and processing units, and other proximal agricultural infrastructure.

Second, positive coefficients of exports of food products and beverages locate the food and beverage export policy within the value chain of agriculture. Thus, upgrading the export policy should encompass the issues of primary production, agro-certification, refrigerated transport, expeditious customs clearance, and agro-export infrastructure towards integration of regional and international markets.

Finally, the agro-industrial policy of Algeria should aim at the enhancement of the food processing industry of value-added goods to reduce the resource burden of the food systems and agro-industrial sector. Active loss of food systems and improve the productivity of resources faster.

The main finding of this study is that food industry production, coupled with food and beverage exports, is positively related to the increase of agriculture in Algeria. Use of ARDL in this context offers insights for formulation of food production policies, but the agro-industrial sector controls will determine the sustainability of the expansion of industrial foods.

This research does have limitations. The model only uses two primary agro-industrial variables. Because of limited data and sample size constraints, it cannot account for rainfall, irrigation, cultivated land, subventions, macroeconomic disturbances, and other variables of this nature.

The analysis could be furthered in future studies where disaggregated food industrial subsectors are used in conjunction with and firm-level data and possibly cross-sectional or other variable data of irrigation, land use, and other variables that represent the environmental pressures and system resilience of the food supply.

7. Policy and Recommendations

Policymakers should prioritize investments in agro-processing facilities proximate to key agricultural zones, reducing transportation costs and encouraging farmers to increase output. Subsidized credit lines and technical assistance programs can enable smallholders to adopt modern inputs and technologies, aligning with the significant positive long-run coefficients.

To capitalize on forward linkages, Algeria must upgrade storage, cold-chain, and transport networks. Establishing public-private partnerships for warehouse development and improving rural road connectivity will mitigate post-harvest losses. Streamlining customs procedures and upgrading export facilities at ports will further boost food and beverage exports, reinforcing the 0.10 % effect on agricultural value added demonstrated.

Robust, real-time data collection on industry and agricultural outputs—building on the publicly available datasets used in this study will enable evidence-based policy adjustments. Developing key performance indicators (KPIs) aligned with the ARDL model's metrics (*VAAG*, *LPBIA*, *LEXAB*) will allow authorities to track progress and identify emerging constraints, ensuring that the food industry reliably translates growth into agricultural development. By implementing these recommendations, Algeria can harness the full potential of its food industry to drive sustainable agricultural growth, enhance food security, and reduce hydrocarbon dependency, fulfilling both sectoral and national economic diversification objectives.

Author Contributions

Conceptualization, S.B. and I.B.; methodology, I.B. and B.A.; software, I.B.; validation, S.B., N.K., and B.A.; formal analysis, I.B. and B.A.; investigation, S.B. and B.A.; resources, N.K. and B.A.; data curation, B.A.; writing—original draft preparation, S.B. and I.B.; writing—review and editing, N.K. and B.A.; visualization, I.B.; supervision, S.B. and N.K.; project administration, I.B.; funding acquisition, I.B. All authors have read and agreed to the published version of the manuscript.

Funding

This work was supported through the Ambitious Funding track by the Deanship of Scientific Research, Vice Presidency for Graduate Studies and Scientific Research, King Faisal University, Saudi Arabia (Grant No.: KF263552).

Data Availability

All data are obtained from the National Office of Statistics, Algeria, a publicly accessible database.

Conflicts of Interest

The authors declare no conflicts of interest.

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